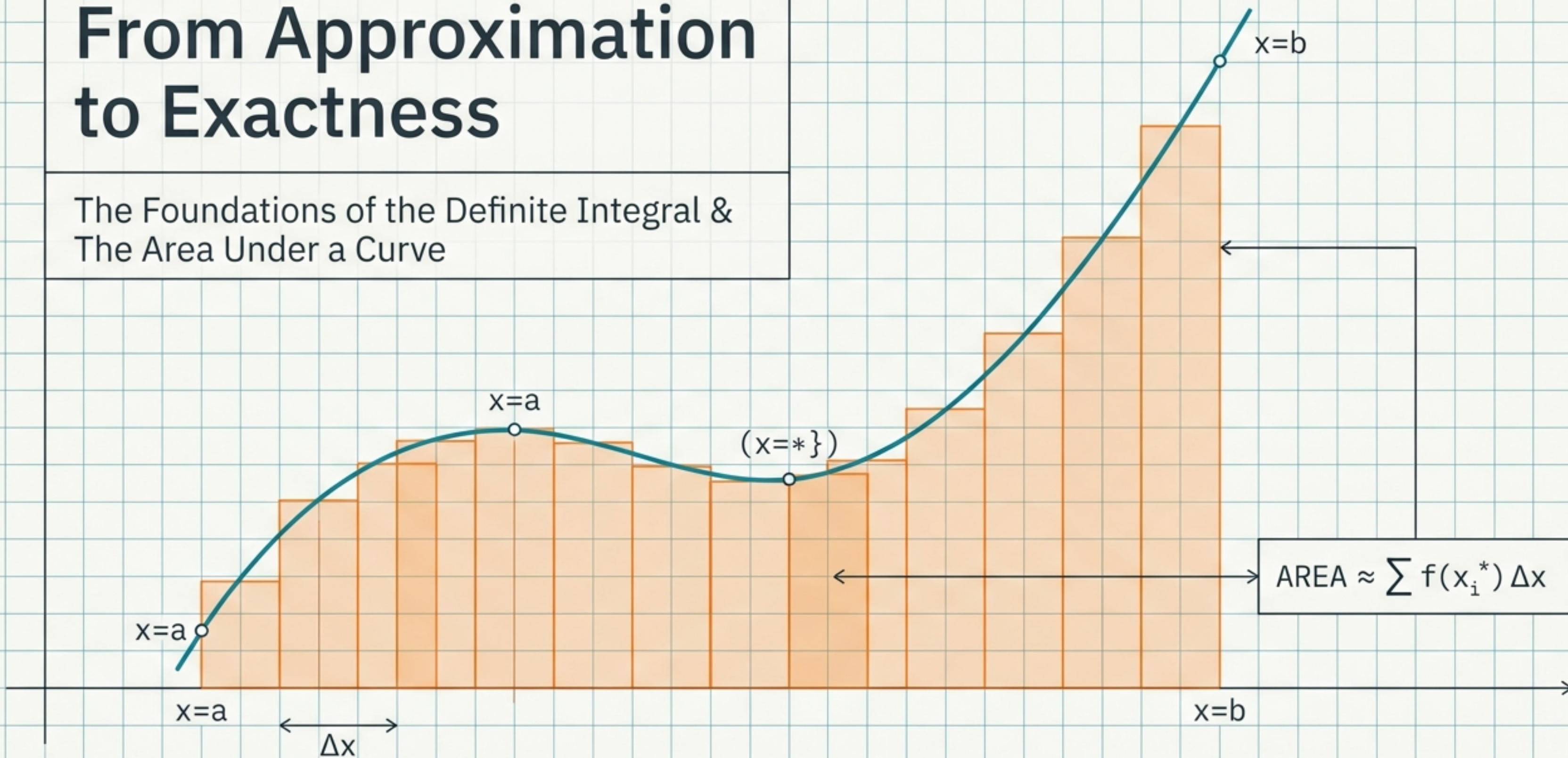
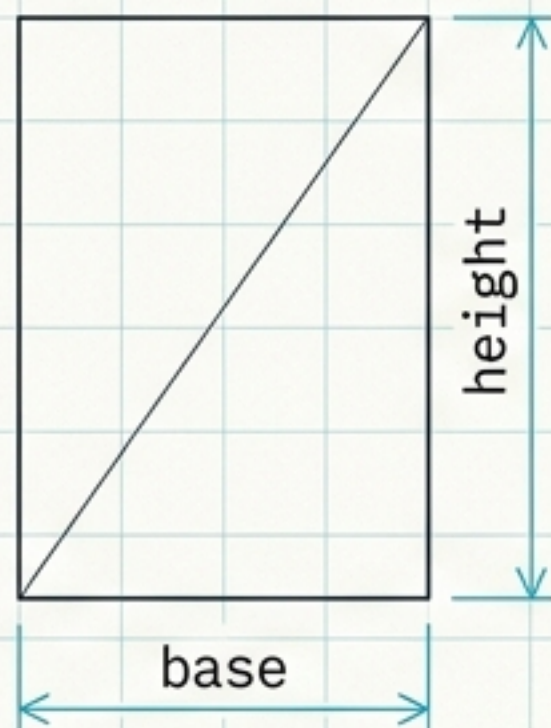


From Approximation to Exactness

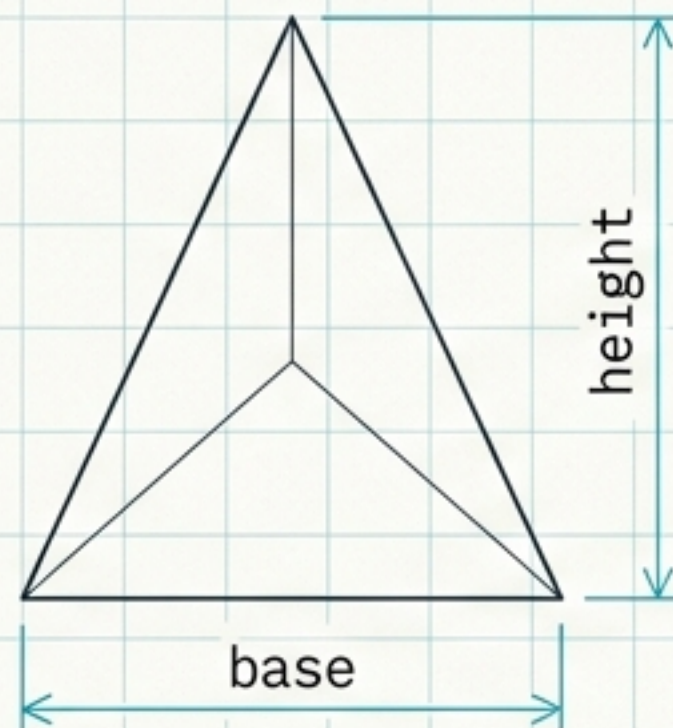
The Foundations of the Definite Integral & The Area Under a Curve



THE KNOWN



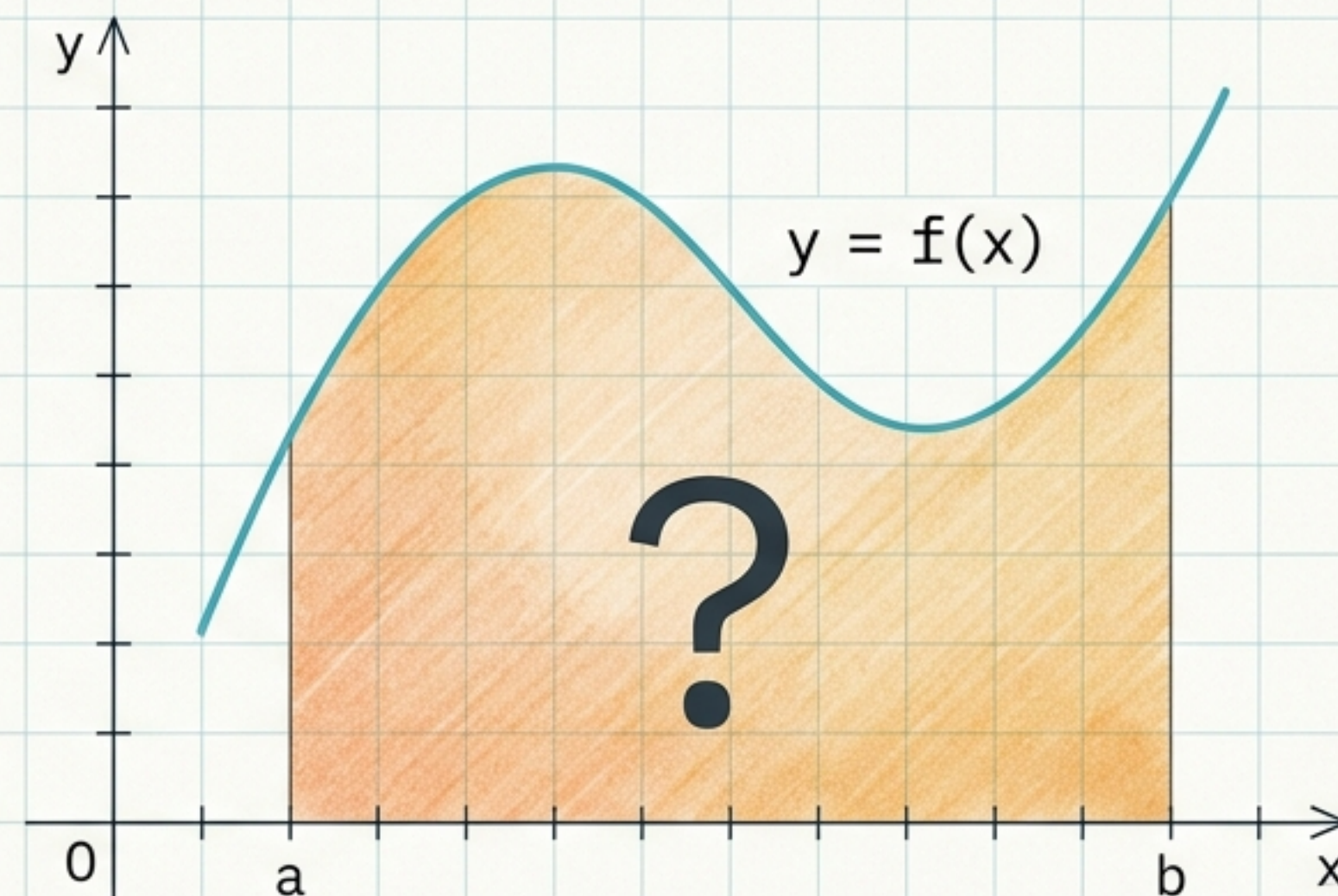
$$A = \text{base} \times \text{height}$$



$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Geometry equips us to measure the perfect and the linear.

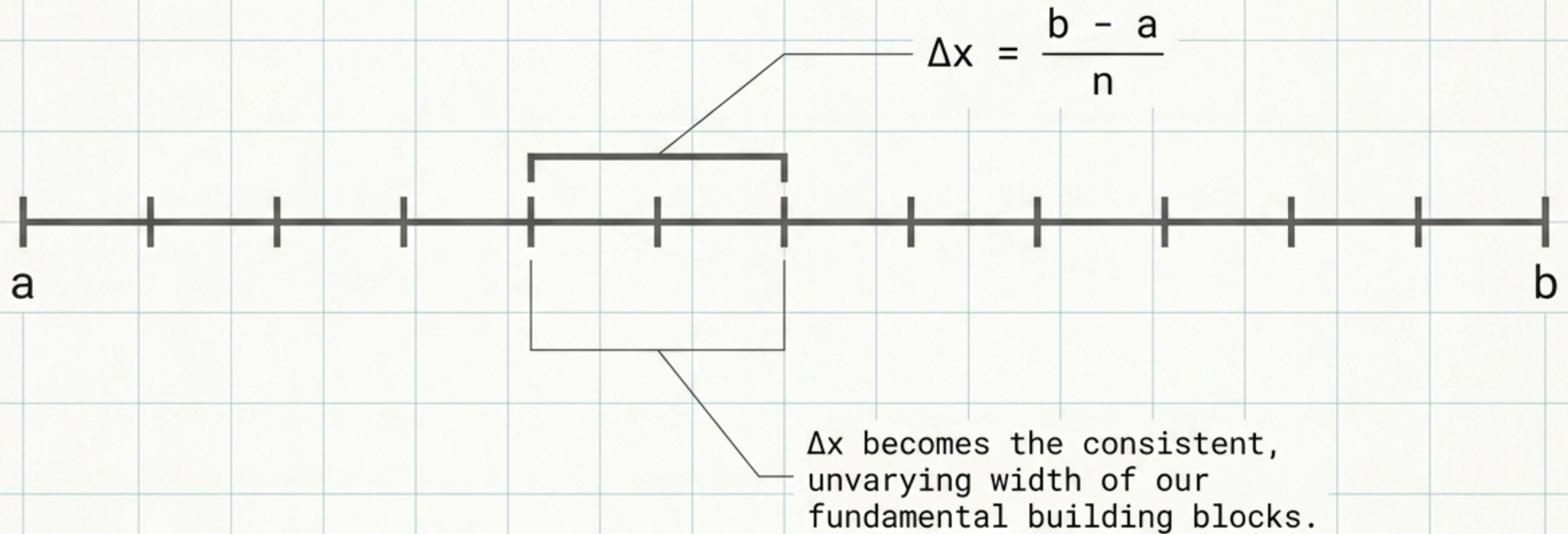
THE UNKNOWN



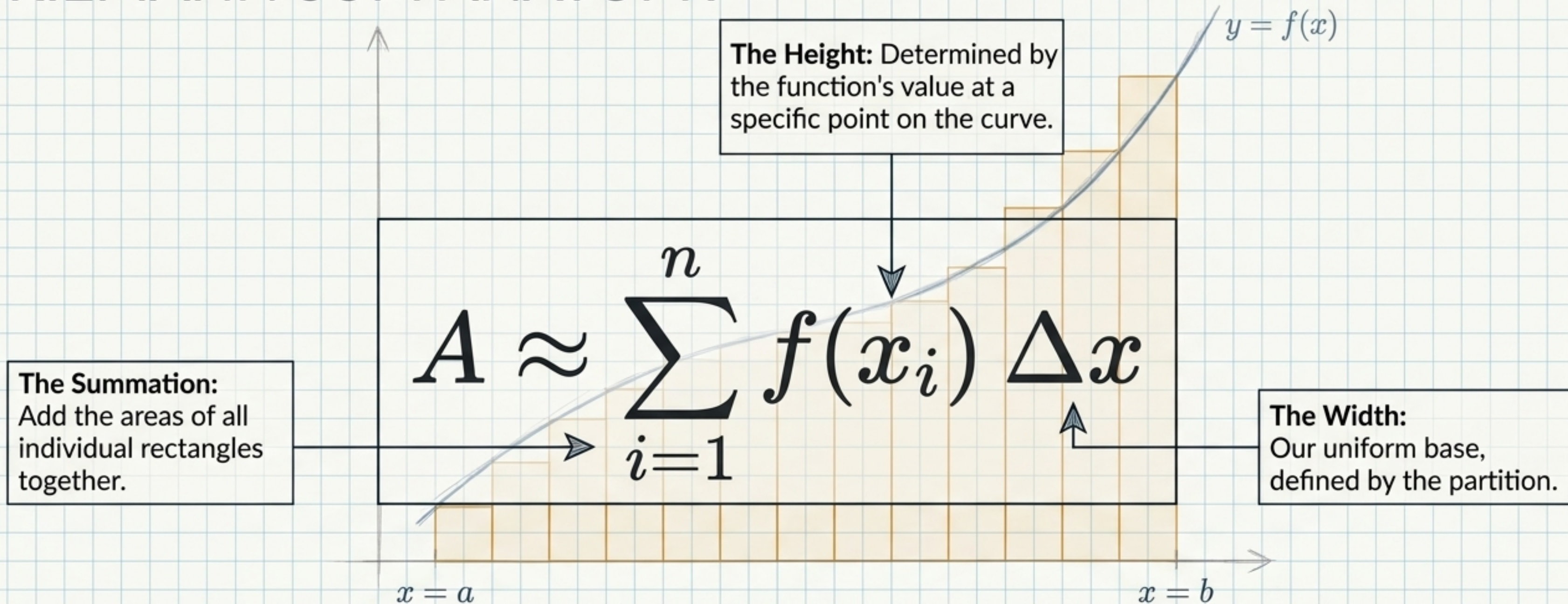
But nature and data are irregular. How do we calculate the exact area of continuous change?

The Uniform Partition

We divide the complex space into n equal, manageable pieces.



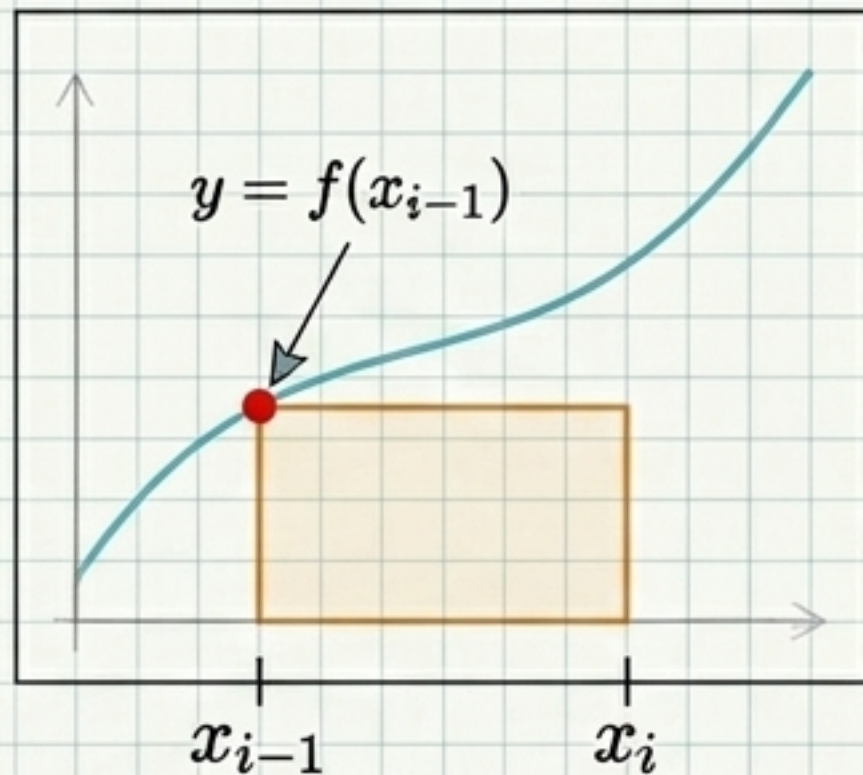
RIEMANN SUM ANATOMY



Area = Sum of (Height $\times$ Width). The Riemann Sum is just elementary geometry operating at scale.

Evaluation Point Matrix

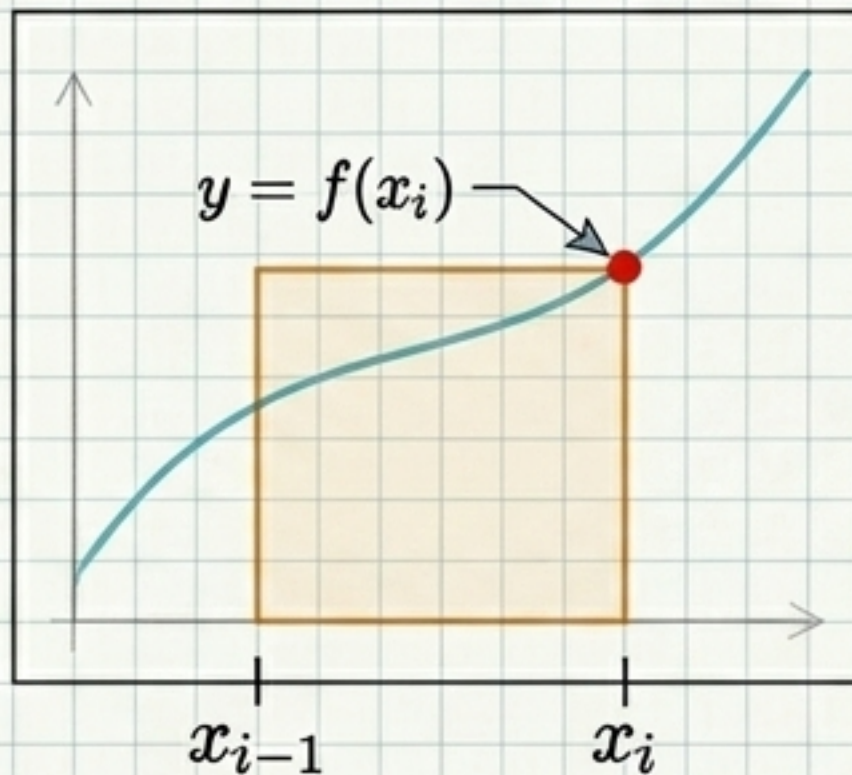
Left Endpoint



$$c_i = x_{i-1}$$

Anchor the height to the start of the sub-interval.

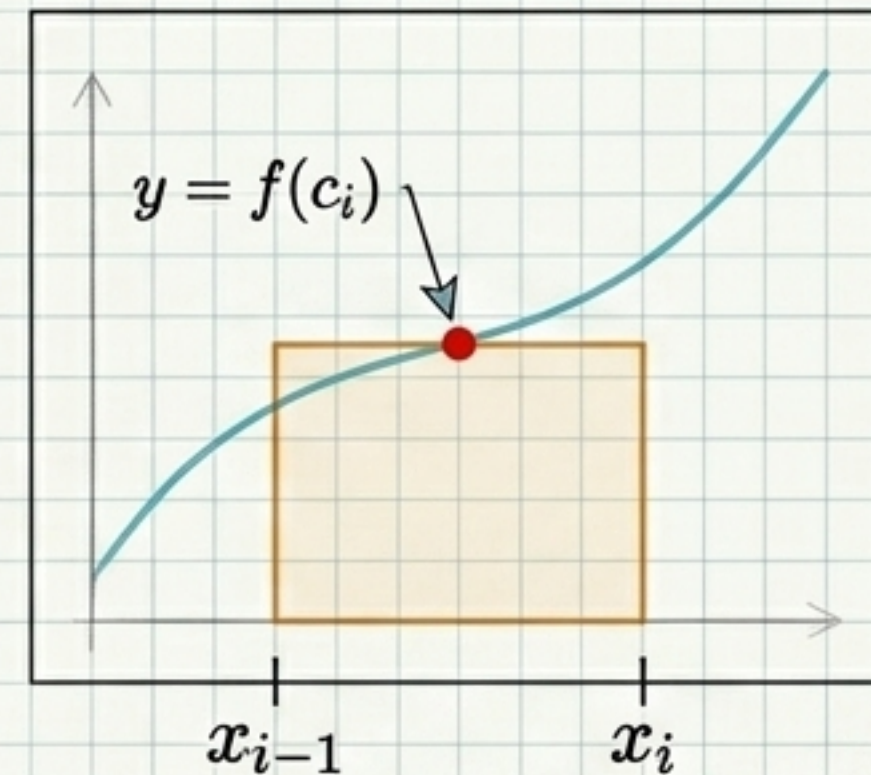
Right Endpoint



$$c_i = x_i$$

Anchor the height to the end of the sub-interval.

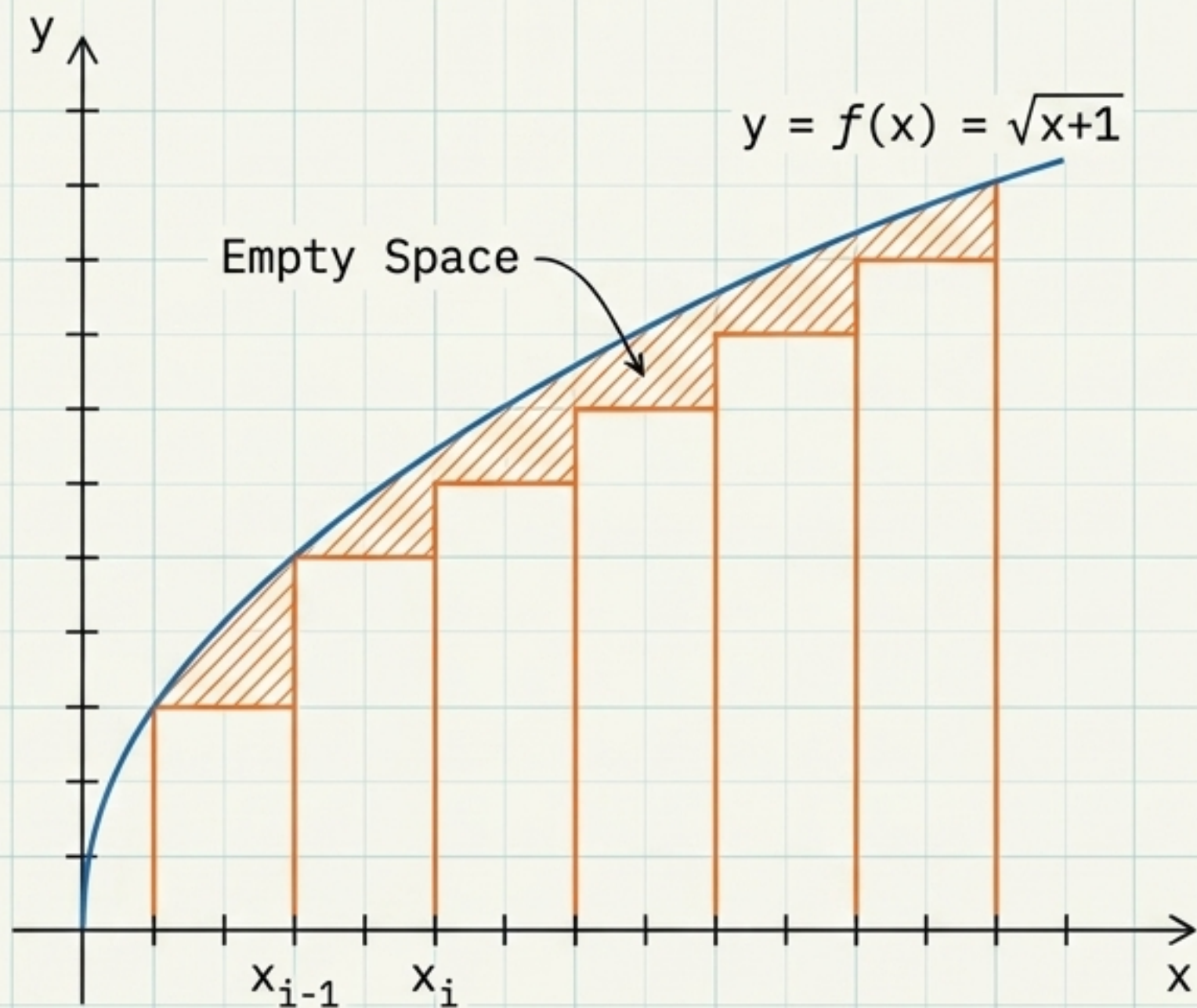
Midpoint



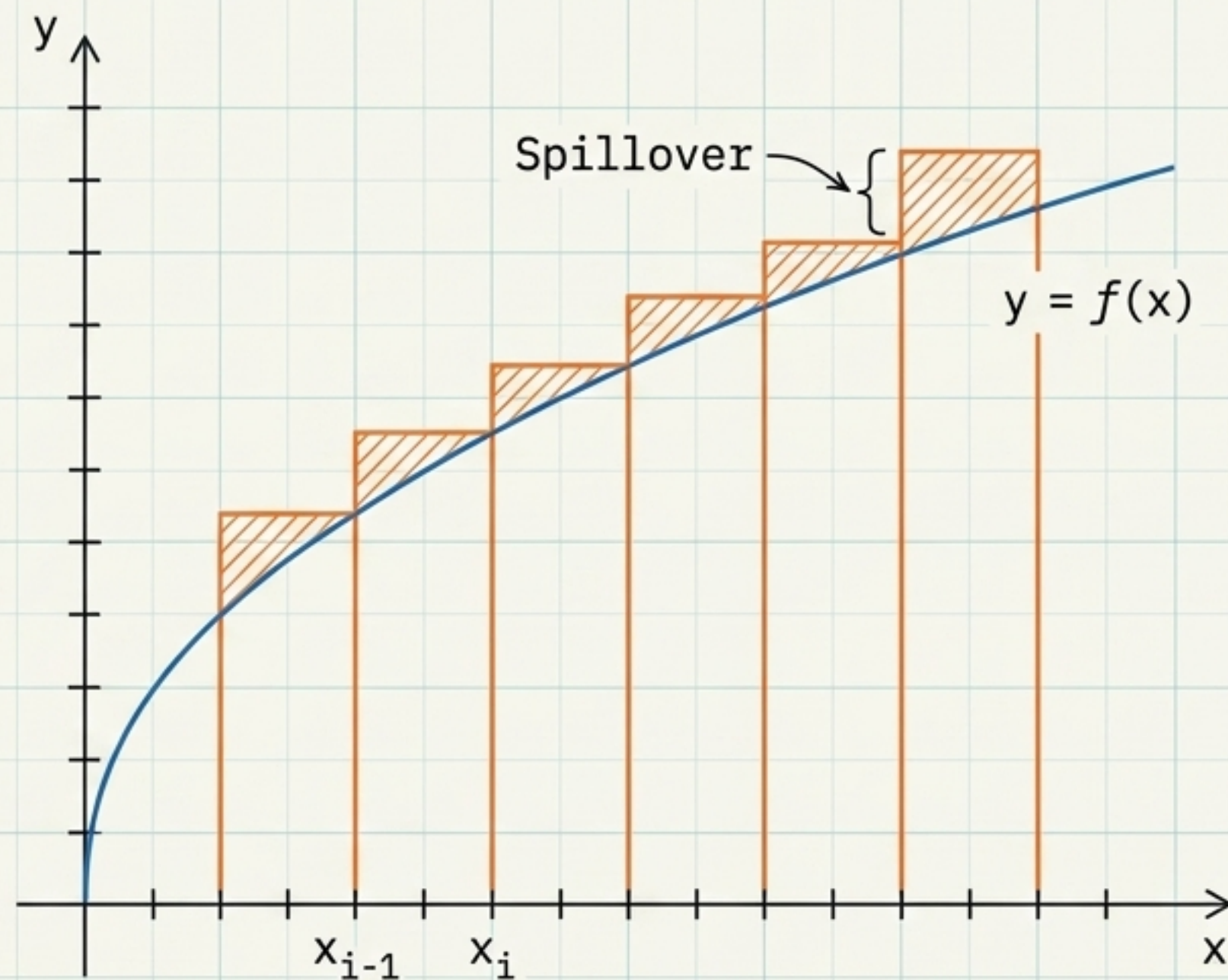
$$c_i = \frac{x_{i-1} + x_i}{2}$$

Average the boundaries. Generates the most mathematically accurate approximation for small values of n .

Underestimation

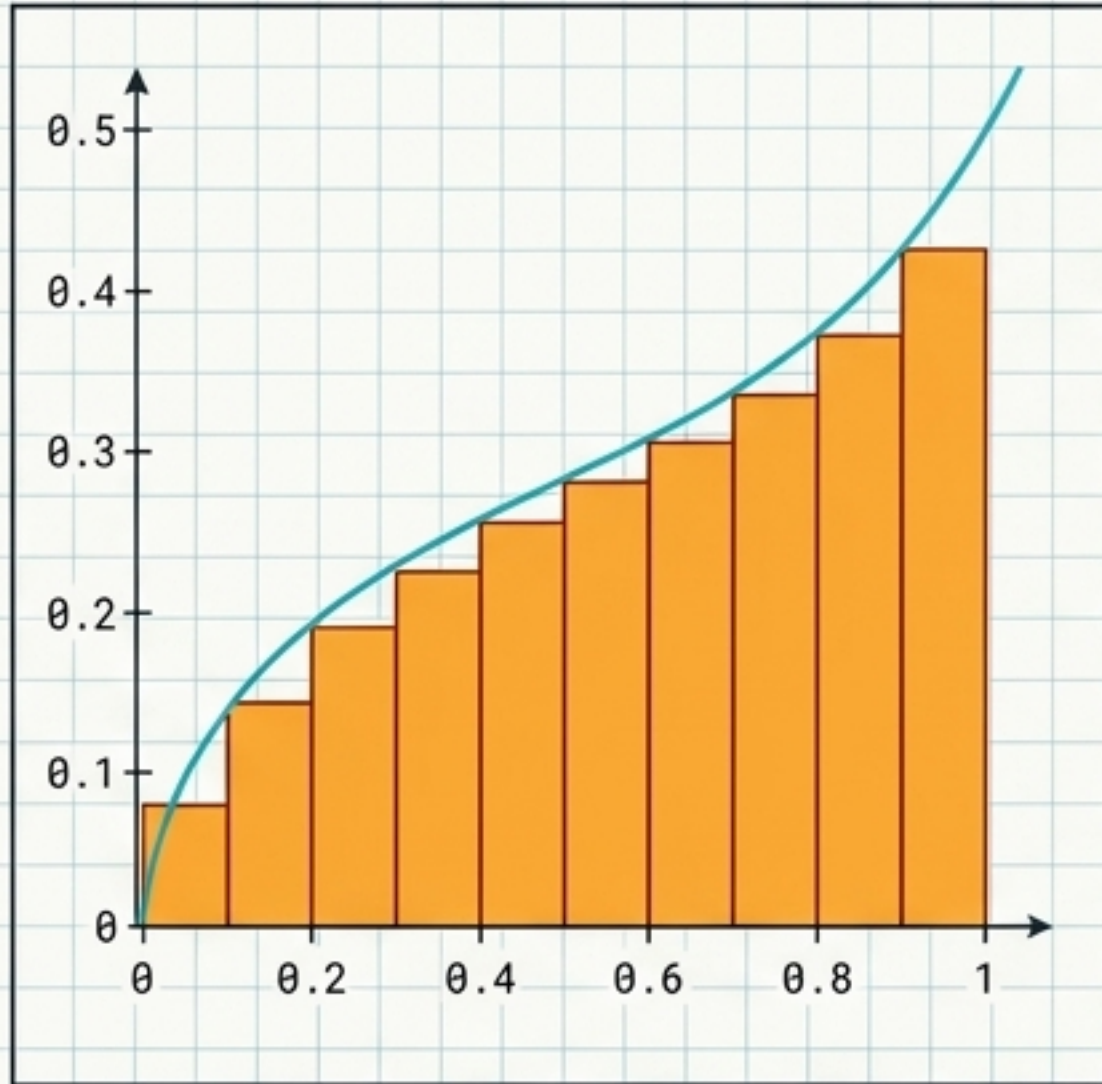


Overestimation

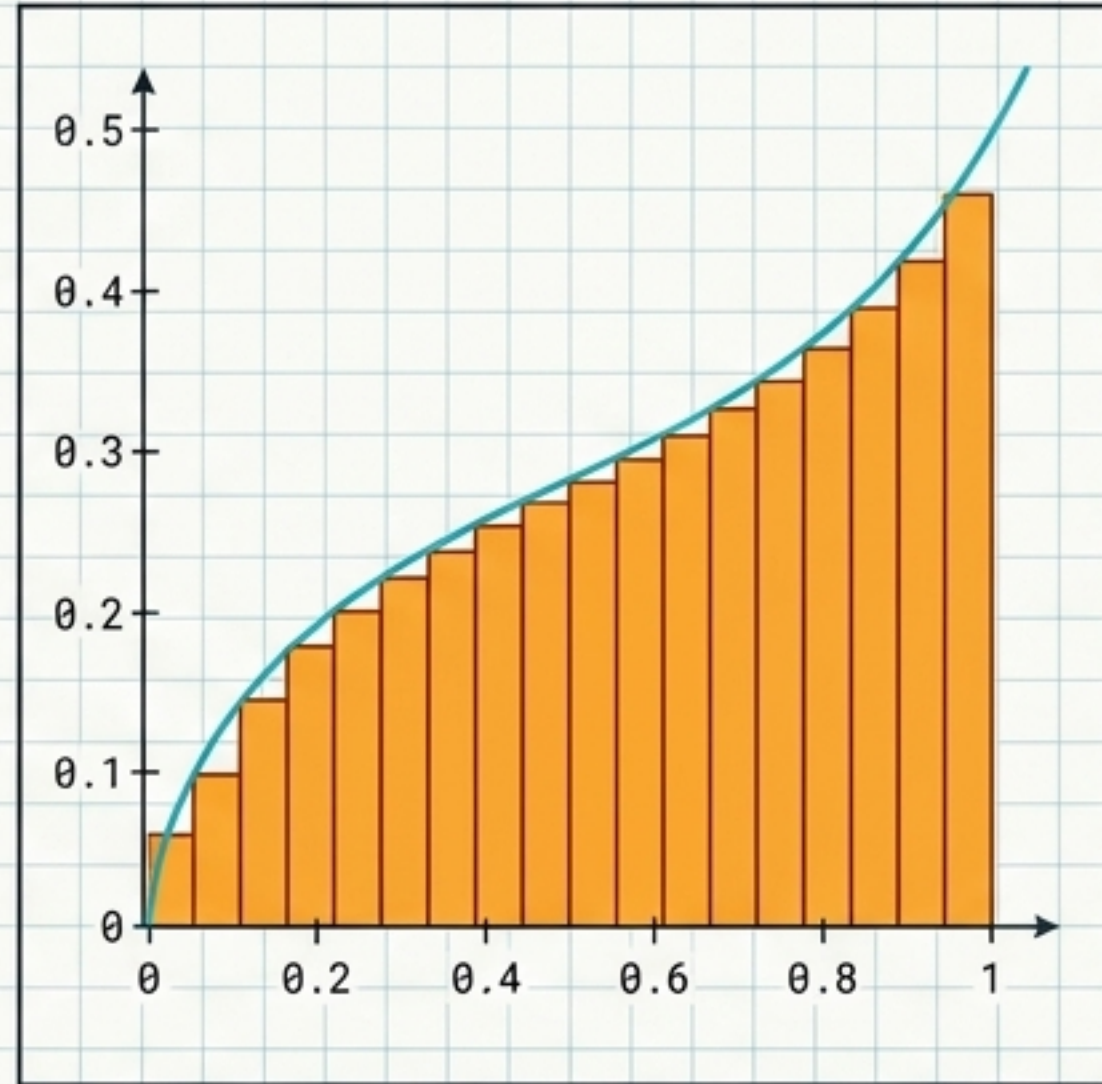


Every finite partition carries a geometric error.
Notice that for a decreasing curve, these behaviors reverse.

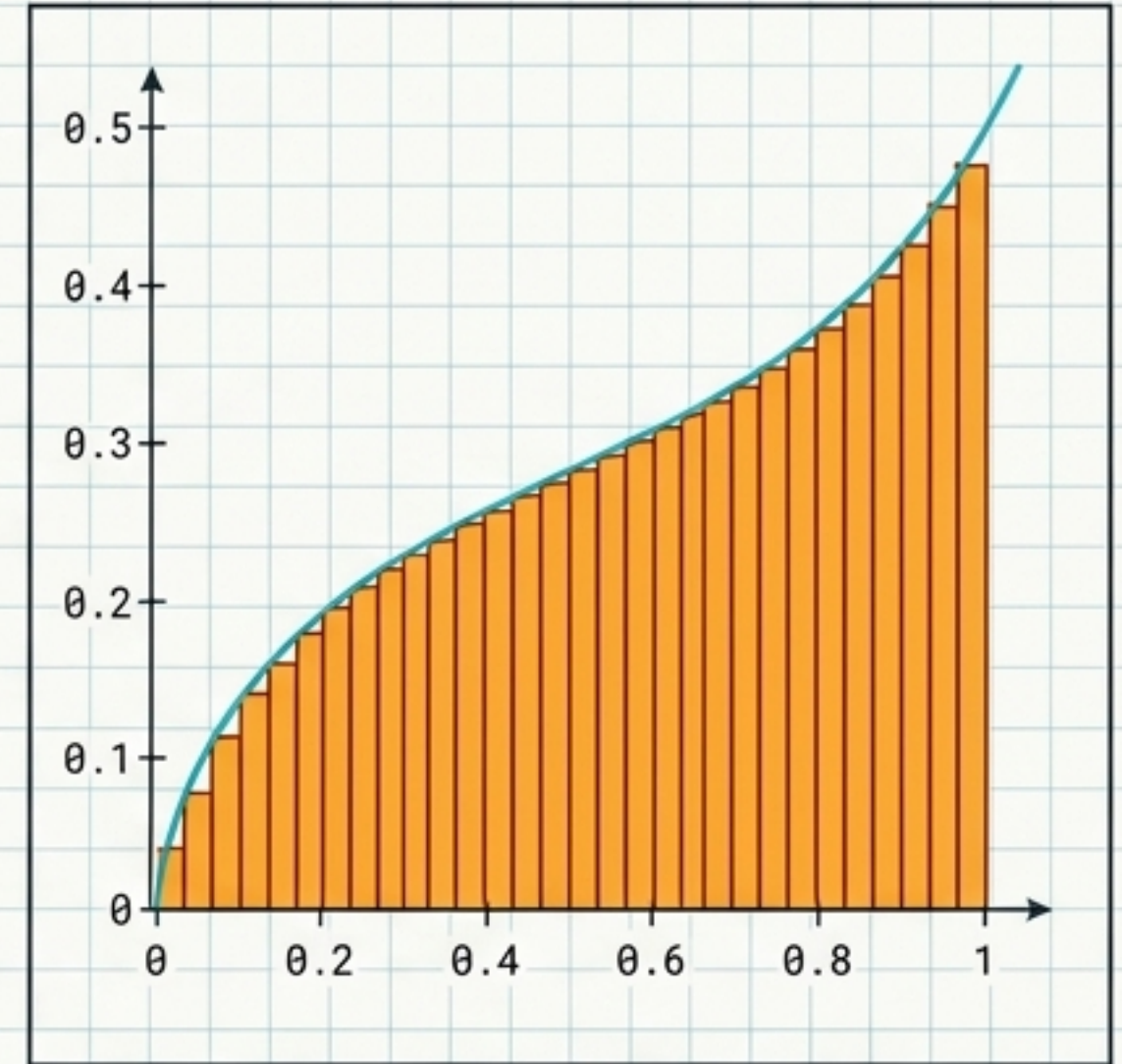
The Progression of Accuracy



$n = 10$



$n = 20$



$n = 40$

As the number of partitions (n) grows, the width (Δx) shrinks. The rigid “staircase” yields to the organic sweep of the curve.

n	Left Endpoint Area	Right Endpoint Area
10	~ 3.38879	~ 3.50595
100	~ 3.44185	~ 3.45357
5000	~ 3.44760	~ 3.44783

Convergence

The **Illusion of Choice**: Though they approach from opposite directions, Left and Right evaluations lock onto an identical geometric truth (≈ 3.44778). The starting rule ceases to matter.

The Leap to Exactness

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x$$

Infinity solves the geometric problem. If we use infinitely many, infinitely thin rectangles ($n \rightarrow \infty$), the error drops to exactly zero. The approximation becomes absolute truth.

The Grand Generalization

Definition 4.1

If f is a continuous function defined on $[a, b]$, the limit of its Riemann sums is defined as **The Definite Integral**.

$$\int_a^b f(x) dx$$

This concept is so fundamental to the mathematics of change that it requires its own architecture and notation.

The Rosetta Stone of Calculus

$$\lim_{n \rightarrow \infty} \sum f(c_i) \Delta x$$

↙ The stylized "S" stands
↘ for continuous Sum.

↙ The function
↘ dictating height.

↙ The infinitesimally
↘ small width.

↙ Lower (a) and
↘ upper (b) bounds.

$$\int_a^b f(x) dx$$

The Definite Integral is simply the ultimate shorthand for an infinite Riemann construction.

Bernhard Riemann: Architect of Integration



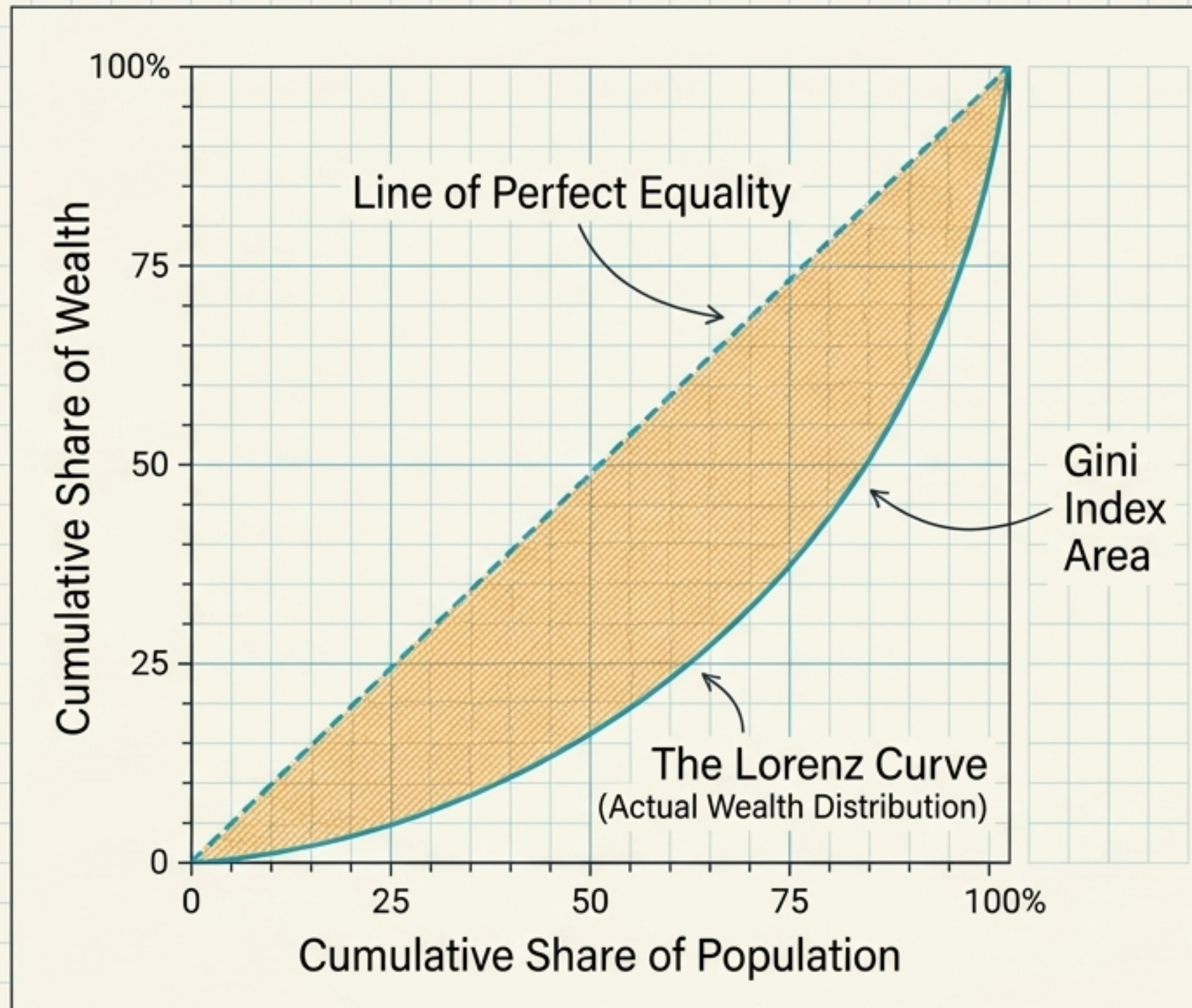
Bernhard Riemann
1826–1866

↙ A German mathematician who formalized the overarching concept of **integration**. ↘

↙ Profoundly influenced modern mathematics by forging deep, unexpected links between analysis and geometry. ↘

↙ His generalized method transformed calculus from intuitive approximations into rigorous, undeniable truths, despite his tragically short life. ↘

Beyond Geometry: The Economics of Inequality



The Problem

Economists measure wealth inequality by comparing an ideal society to the real distribution of wealth (The Lorenz Curve).

The Solution

The Gini Index is calculated by finding the exact area between the line of perfect equality and the curved Lorenz line.

The Takeaway

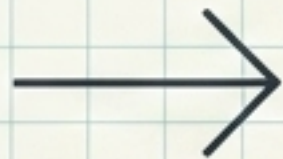
The Definite Integral makes measuring the health of a global economy mathematically possible.

1. Geometry (The Partition)



Δx

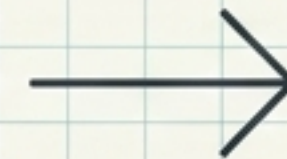
Slicing continuous space into discrete, manageable blocks.



2. Algebra (The Summation)



Systematically calculating and adding the area of finite rectangles (Riemann Sums).



3. Calculus (The Integral)



Applying the limit of infinity to collapse the error space to zero, revealing the exact truth.

Calculus is the mathematics of continuous change, built upon the foundation of infinite discrete constructions.